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Managing unreliable suppliers under varying risk preferences: all-or-nothing versus fractional delivery guarantees

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Abstract
This paper revisits the classical newsvendor problem with unreliable suppliers, examining both risk-neutral and risk-averse formulations. We study multi-sourcing decisions under yield uncertainty, distinguishing between *all-or-nothing* (AON) suppliers, who either deliver fully or fail completely, and suppliers offering *fractional delivery guarantees* (FDGs) that ensure partial fulfillment under disruption. For the risk-neutral case, we show that the supplier ranking rule proposed previously, based on *effective cost*, is not optimal. Instead, we prove a cost-ordered dominance property: suppliers are considered in nondecreasing unit procurement cost (a higher-cost supplier cannot be active if a lower-cost supplier is inactive), while disruption risk affects the optimal quantities and the activation thresholds within this structure. Fractional guarantees generate nonlinear and potentially unimodal patterns in order quantities and service levels, leading to different activation behavior compared with AON models. In the risk-averse case, we extend the mean–variance formulation and demonstrate that an earlier study omitted part of the profit variability, resulting in an underestimation of risk. We derive the complete mean–variance expression and introduce a simplified weighted-variance formulation that captures risk more accurately while remaining computationally efficient. The results show that minimizing weighted variance activates all suppliers when stockout penalties are present, whereas minimizing full variance offers greater protection when reliability is low. Overall, our study offers clearer guidance for supplier selection and ordering decisions under uncertainty and risk aversion, providing new analytical and numerical insights into risk-averse supplier portfolio management.

Keywords Production · Supplier selection · Mean–variance · Newsvendor model

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1 Introduction

Understanding and mitigating supply chain uncertainties is crucial for developing effective inventory strategies. One such strategy, *risk diversification*, involves spreading risk across multiple suppliers to reduce dependence on any single source. This approach is particularly relevant when facing unreliable suppliers, categorized into capacity uncertainty, major disruptions, and yield uncertainty (Mohammadivojdan et al. 2022). This study explores yield uncertainty in a newsvendor setting with multiple sourcing options. It examines two types of suppliers: *all-or-nothing (AON) suppliers*, who either fulfill an order in full or fail completely (Mohammadivojdan et al. 2022), and *fractional delivery (FD) suppliers*, who may deliver only a portion of the ordered quantity (Ray and Jenamani 2016a, b). In an AON model, supplier failure results in unmet demand, increasing risk. In contrast, a fractional delivery model reduces this risk by ensuring partial fulfillment but adds complexity to decision-making.

This study aims to analyze the newsvendor problem in the presence of unreliable suppliers and develop tailored strategies for both risk-neutral and risk-averse newsvendors. For the risk neutral setting where suppliers provide *fractional delivery guarantee* (FDG), Ray and Jenamani (2016b) propose an ordering method based on the measure *effective cost* (that depends on risk and cost parameters) in the risk-neutral setting. We show that prioritizing suppliers according to this measure is not optimal. We prove that in the optimal solution suppliers must be ordered based on *unit procurement cost*. That is, disruption risk cannot overturn the cost-based activation order (a higher-cost supplier cannot be active while a lower-cost supplier is inactive), although shipment guarantee and disruption probability do affect the optimal order quantities and the ranges of disruption probability in which additional (higher-cost) suppliers become active. Note that although this is also in line with *cost-first, reliability-second* insight from the risk-neutral supplier selection literature (Dada et al. 2007), ours is the first study which proves this result with the FDG setting.

The difference between AON and FDG significantly influences procurement strategy, yet the impact of FDG remains largely unexplored in the existing literature. We show that a key distinction between these settings lies in ordering strategies and service level behavior. In a single-sourcing scenario, the AON model maintains a fixed order quantity, following the classical newsvendor solution, while service levels decline as the failure probability of the preferred supplier increases. In contrast, in the FDG setting, the optimal order quantity increases in probability of disruption, while the service level exhibits a pattern of decrease–then–increase (proven for uniform demand and demonstrated numerically for other distributions). This affects the supplier activation dynamics when multiple suppliers exist: in the AON setting, higher disruption probability of the favorable supplier does not lead to the elimination of other active suppliers, whereas in the fractional setting, it does. For a fixed disruption probability, a higher shipment guarantee increases the service level. As the shipment guarantee improves, the optimal order quantity follows a unimodal pattern: it coincides with the classical newsvendor solution, increases initially, then decreases, and ultimately converges back to the classical newsvendor solution (proven for a broad class of demand distributions). Furthermore, we find that when the lowest-cost sup-

plier provides a minimum yield guarantee under disruption, a threshold determined by the unit costs of competing suppliers, this supplier is always selected as the single source regardless of its disruption risk (proven for uniform demand).

In the risk-averse setting, we adopt a mean–variance framework similar to Ray and Jenamani (2016a), where suppliers provide fractional delivery guarantees (FDGs). However, their formulation overlooks critical components of profit variability: specifically, scenario-driven fluctuations in expected profits. This omission leads to a systematic underestimation of risk and a less conservative decision framework. We refer to this new risk measure as the *weighted variance*. We analyze the differences between these risk measures and find that weighted variance is a more computationally efficient risk measure. When the stockout cost (loss of goodwill penalty) is zero, the weighted variance objective in a single unreliable supplier setting increases with order quantity, while variance exhibits a unimodal behavior, initially increasing and then decreasing. When stockout cost increases, the weighted variance function becomes unimodal, while the variance function becomes more complex.

Minimizing variance, rather than weighted variance, can offer additional advantages when stockout costs are high and supplier reliability varies significantly. This is particularly relevant for AON suppliers with a moderate likelihood of disruption, where uncertainty is at its peak. We find that for such suppliers, when disruption risk is very high, both weighted variance and variance become small because the supplier rarely delivers, resulting in more predictable, albeit lower, outcomes. This suggests that in a highly risk-averse framework, a supplier with a very high risk of disruption may still be preferable in extreme cases, as its minimal contribution helps stabilize overall profit variability. This highlights the significance of mean–variance settings, where the low expected profit of such suppliers typically renders them undesirable. Furthermore, Ray and Jenamani (2016a) propose an approach that requires solving the Mean–variance problem, using weighted variance as a risk measure, optimally for a subset of suppliers with a nonlinear solver. However, when most suppliers are active, this approach can be as computationally intensive as solving the original problem. This study introduces a new, computationally efficient algorithm based on optimality conditions.

Our analytical results and the numerical study lead to various new insights on risk-averse supplier portfolio management. When the decision maker minimizes the weighted variance, we analytically show that no supplier is eliminated in the optimal solution when stockout costs exist for a large class of demand distributions. We also demonstrate numerically that this result holds for the variance-minimizing newsvendor model. While the risk-minimizing newsvendor tends to order from all suppliers, our findings indicate that the supplier acceptance decisions, following the preference order of a risk-neutral setting, remain primarily driven by procurement cost within the mean-risk framework. We also demonstrate that, while a risk-averse decision-maker prefers to order from all suppliers, the allocation of order quantities is largely influenced by the variance of the reliability random variable. Finally, Ray and Jenamani (2016a) claim that a risk-neutral newsvendor orders more than a risk-averse one. However, we show that this generalization does not always hold when the suppliers are highly unreliable and when stockout costs are high, aligning with the conclusions of Fleuren (2025).

To summarize, this study makes several key contributions. (i) For the risk-neutral case with fractional delivery guarantees (FDG), we show that the effective-cost–based supplier ranking proposed by Ray and Jenamani (2016b) does not yield the optimal ordering strategy; instead, we prove that suppliers must be prioritized solely by unit procurement cost, thus formalizing the "cost-first, reliability-second" principle in the FDG context. (ii) We characterize ordering and service-level behavior under unreliable sourcing and reveal a fundamental contrast between all-or-nothing (AON) and FDG settings: while AON yields monotone service level decline with rising failure risk, FDG results in non-monotonic (unimodal) patterns. (iii) For the risk-averse setting, we identify a structural omission in the variance formulation of Ray and Jenamani (2016a) and introduce a new measure, *weighted variance*, that captures scenario-driven variability. We analytically compare *variance* and *weighted variance*, establishing when each yields more conservative procurement and deriving conditions under which even highly unreliable suppliers remain active due to stability effects. (iv) We develop a computationally efficient algorithm for the selection of Mean–variance suppliers under FDG, using optimality conditions to avoid solving nonlinear subproblems, thereby improving scalability. (v) Across both analytical results and extensive numerical experiments, we demonstrate that risk-averse newsvendors typically diversify across all suppliers, yet allocation magnitudes remain primarily cost-driven, and we refine the statement in Ray and Jenamani (2016a) that risk aversion invariably reduces order quantities. We show that this conclusion does not hold universally and identify settings in which the opposite behavior arises, in line with Fleuren (2025).

The remainder of the study is organized as follows. First, in Sect. 2, we review the relevant literature. In Sect. 3, we introduce the standard newsvendor problem with a reliable supplier and present the Mean–variance analysis. Section 4 explores the risk-neutral newsvendor problem with unreliable suppliers. The risk-averse setting is then examined in Sect. 5. In Sect. 6, we present the results of the sensitivity analysis to derive managerial insights. Finally, concluding remarks and recommendations are provided in Sect. 7.

2 Literature review

The literature on the newsvendor problem is extensive, including early theoretical frameworks as well as complex models that account for various sources of uncertainty. Yield risk in inventory systems has also been explored across diverse contexts, and readers are encouraged to consult the reviews by Yano and Lee (1995), Grosfeld-Nir and Gerchak (2004), Vakharia et al. (2009), along with the references they cite.

The research considering unreliable suppliers is categorized by Mohammadivojdan et al. (2022) into three categories: yield uncertainty, capacity uncertainty, and major disruptions. Anupindi and Akella (1993) studied both single- and multi-period problems with uncertain demand in the context of unreliable suppliers and found that ordering from the supplier with the lowest procurement cost is generally optimal. Agrawal and Nahmias (1997) studied supplier selection in scenarios with either identical or diverse suppliers, assuming deterministic demand and yield uncertainty. Their

findings indicate that rationalizing the supplier base can reduce costs but increases the risk of supply disruptions. Dada et al. (2007) show that if payment is made at delivery, cost is an order qualifier and reliability an order winner. Mohammadivojdan et al. (2022) supported this latter view in settings with AON suppliers in the presence of a backup supplier or spot markets. Since a supplier may be unwilling to commit production capacity without a guaranteed revenue, the newsvendor pays an additional nonrefundable production cost for each unit ordered in Mohammadivojdan et al. (2022). In this case, it is demonstrated that suppliers should be prioritized based on the ratio of unit cost plus production cost to expected yield (or availability probability). This result aligns with Federgruen and Yang (2009), who addressed a similar problem involving general yield uncertainty but restricted the payment structure to include only the production cost. Merzifonluoglu and Feng (2013) studied the newsvendor problem with unreliable suppliers and found that, when fixed ordering costs and supply capacity exist, cost as order qualifier and reliability as order winner do not hold. Ray and Jenamani (2016b) investigated the newsvendor problem with multiple unreliable suppliers, where each supplier faces a probability of failure but guarantees delivery of a fixed percentage of the order quantity even in the case of failure. While their payment structure assumes payment upon delivery (excluding production costs), they argued that supplier selection should be based on a novel measure, *effective costs*, which incorporates reliability. We examine the same problem and demonstrate that, even when suppliers provide FDGs, supplier selection should still be driven by the unit order cost.

We next review the related literature on the Mean–variance framework, with Chiu and Choi (2016) providing a comprehensive overview of its application in supply chain risk. Chen and Federgruen (2000) analyzed several fundamental inventory models using a Mean–variance approach, modeling a quadratic utility function for the decision-maker. They study the newsvendor problem with cost and profit models, demonstrating that variance increases with order quantity. However, when stockout cost is considered, variance is no longer a monotonically increasing function, as discussed by Wu et al. (2009). Wu et al. (2009) examined the classical newsvendor problem within a Mean–variance framework, deriving structural properties of the optimal order quantity under power-distributed demand. Fleuren et al. (2026), with a full exposition also provided in the doctoral thesis Fleuren (2025), study the same problem and derives a *critical-fractile-style* optimality condition analogous to the classical risk-neutral benchmark, establishing the uniqueness of the solution for a broad class of demand distributions. Xue et al. (2016) studied the newsvendor problem with multiple unreliable suppliers within the Mean–variance framework, incorporating general yield uncertainty. Their model assumes payments are made at the time of ordering (not at the delivery), considering only production costs. By approximating leftover inventory with a normal distribution, they demonstrated that active suppliers are selected similarly to the risk-neutral setting, in ascending order of their effective marginal cost (production cost relative to expected yield), particularly under high demand uncertainty. A numerical study further confirmed this pattern under low demand uncertainty.

Ray and Jenamani (2016a) extended Ray and Jenamani (2016b) by analyzing the newsvendor problem with multiple unreliable suppliers from a risk-averse perspec-

tive. While their study employs the mean–variance framework, it primarily focuses on weighted variance, which captures the expected variance of the newsvendor's profit under supplier disruption scenarios. In this study, we compare the mean-weighted variance objective with the Mean–variance objective and find that the latter places greater emphasis on diversification, leading to the selection of more suppliers. Interestingly, our numerical results confirm that while supplier activation for a risk-averse newsvendor can be based on procurement cost, order quantities are influenced by risk aversion. Ray and Jenamani (2016a) propose a procedure that requires solving the mean-weighted problem to optimality for a subset of suppliers using a nonlinear solver—an approach that can be as computationally demanding as solving the original problem, or even more so when most suppliers are active. We propose a solution algorithm based on optimality conditions that, across various numerical instances, performs on par with a commercial nonlinear solver for the risk-averse case. Finally, while Ray and Jenamani (2016a) claim that a risk-neutral decision-maker would always order more than a risk-averse one, we demonstrate that this result does not hold when high stockout costs are considered and suppliers are unreliable.

3 Preliminaries: framework for the newsvendor problem

This section outlines the key notation for the risk-neutral and risk-averse newsvendor problem. In this context, a retailer, referred to as the newsvendor, orders Q units of a short-life-cycle product prior to the realization of demand, denoted by D. Demand is modeled as a continuous random variable with cumulative distribution function (cdf) $F(\cdot)$ and probability density function (pdf) $f(\cdot)$, mean μ, and variance σ^2, supported on an interval $[a, b]$, where a and/or b may be infinite (e.g., under Normal demand). Any unsold inventory can be salvaged at a price r per unit, which is lower than the procurement cost c. Conversely, if the demand D exceeds the available stock Q, a penalty cost k is incurred for each unit of unmet demand due to the loss of goodwill. The unit selling price is denoted by s, and to avoid unrealistic and trivial cases, we assume $s > c > r$ and $k > 0$. Let $G(Q; D)$ denote the profit of the newsvendor ordering Q and facing demand D, then:

$$G(Q;D) = s\min\{D,Q\} - cQ + r(Q-D)^+ - k(D-Q)^+ \tag{1}$$

Letting $\ell(Q) = E\left[(D-Q)^+\right]$ denote the (first order) loss function and $n(Q) = E\left[(Q-D)^+\right]$ denote the (first order) complementary loss function, the expected profit function could be written as

$$G_E(Q) = (s-c)Q + (r-s)n(Q) - k\ell(Q). \tag{2}$$

The optimal order quantity for the risk-neutral newsvendor is given by $Q^* = F^{-1}\left(\frac{s+k-c}{s+k-r}\right)$. We next formulate the newsvendor problem using the mean–variance objective (see Wu et al. 2009):

$$\max_{Q \geq 0} G_E(Q) - \alpha G_V(Q). \tag{3}$$

where $G_V(Q) = \text{Var}[G(D;Q)]$ represents the variance of the profit, and $\alpha \geq 0$ quantifies the level of risk aversion ($\alpha = 0$ corresponds to the risk-neutral newsvendor). We can formulate the variance:

$$G_V(Q) = (s-r)^2 n_2(Q) + k^2 \ell_2(Q) - \left[(s-r)n(Q) + k\ell(Q)\right]^2, \tag{4}$$

where $n_2(Q) = E[((Q-D)^+)^2]$ is the second order complementary loss function and $\ell_2(Q) = E[((D-Q)^2)]$ is the second order loss function (see Fleuren 2025 for the derivation of the variance). When $k = 0$, this function increases with Q. However, when $k > 0$, it is not possible to conclude the concavity or convexity of the function $G_V(Q)$. Therefore, we cannot guarantee that the stationary point indicates a unique minimum or maximum based on convexity alone. Fleuren (2025) demonstrated that the first-order conditions provide a global optimal solution for certain demand classes such as symmetric demand distributions with an increasing hazard rate (IHR). This class includes various symmetric distributions, such as the normal, continuous uniform, symmetric triangular, logistic, Laplace (double exponential), and beta (with parameters $\alpha = \beta \geq 0.8$) distributions (see, e.g., Rinne 2014 and Merzifonluoglu and Geunes 2025). Moreover, for any demand distribution with IHR and a decreasing reversed hazard rate (DRHR) (including Weibull, gamma, Pareto, and log-normal distributions), the first-order conditions also yield a global optimum. Similarly, we cannot conclude that the Mean–variance objective is concave or convex. However, Fleuren (2025) established that a unique optimal solution exists, which lies between risk-neutral and risk-averse solutions for the demand classes mentioned above which we also consider in this study.

4 Risk-neutral newsvendor with unreliable suppliers

This section examines a variant of the newsvendor problem where the newsvendor has the option to order from multiple unreliable suppliers. The objective is to determine the optimal order quantity from each supplier, Q_i, given supplier-specific unit procurement costs, c_i, and disruption risk under which only a fraction of the ordered quantity may be delivered. The risk of partial delivery from supplier i is modeled as a Bernoulli random variable, B_i, with a disruption probability p_i. The percentage of the order quantity that supplier i can still deliver in the event of a disruption (referred to as the *yield under risk* or *FDG*) is denoted by y_i, where $0 \leq y_i \leq 1$. This framework accommodates suppliers with complete failure ($y_i = 0$) as well as perfectly reliable suppliers ($y_i = 1$). Therefore, the quantity received by the newsvendor is $Q_i R_i$, where R_i represents the reliability factor given by $R_i = 1 - (1 - y_i)B_i$. $E(R_i)$ denotes the expected supply received per unit order placed with supplier i, given by: $E(R_i) = 1 - (1 - y_i)p_i$. The variance of the reliability factor is: $\text{Var}(R_i) = (1 - y_i)^2 p_i(1 - p_i)$. We assume that the disruption events affecting the suppliers and the demand realizations are independent.

To clarify the notation and improve the explanation, we introduce a *disruption scenario* u, which uniquely represents the failure states of the suppliers through the vector $\{b_{1u}, b_{2u}, \ldots, b_{nu}\}$, where:

$$b_{iu} = \begin{cases} 1 & \text{if supplier } i \text{ experiences a disruption in scenario } u, \\ 0 & \text{otherwise.} \end{cases}$$

Each scenario[1] occurs with probability $P_u = \prod_{i=1}^{n} [b_{iu}p_i + (1 - b_{iu})(1 - p_i)]$ and there are 2^n scenarios. Let $m_{iu} = Q_i(1 - (1 - y_i)b_{iu})$ denote the quantity received from supplier i in scenario u. Then, the total quantity received in scenario u is given by $Q_u = Q_u = \sum_{i=1}^{n} m_{iu}$. Let $G(Q_u; D) = \sum_{i=1}^{n}(s - c_i + k)m_{iu} - kD - (s - c_i + k)(Q_u - D)^+$ denote the profit function for a particular scenario u and customer demand D, and $G_E(Q_u) = E[G(Q_u; D)] = \sum_{i=1}^{n}(s - c_i + k)m_{iu} - k\mu - (s - r + k)n(Q_u)$ denote the expected profit given disruption scenario u occurs. Then, the expected profit becomes

$$\begin{aligned} G_E(Q) &= \sum_{u \in U} P_u E[G(Q_u; D)] \\ &= \sum_{u \in U} P_u \left(\sum_{i=1}^{n} (s - c_i + k)m_{iu} - k\mu - (s - r + k)n(Q_u) \right), \end{aligned} \tag{5}$$

The objective of a risk-neutral newsvendor is to maximize expected profit over $Q_i \geq 0,\ i = 1, \ldots, n$. Under standard regularity conditions, the Karush–Kuhn–Tucker (KKT) conditions provide the corresponding first-order optimality conditions. For supplier i, define the critical ratio as $CR_i := \frac{s - c_i + k}{s - r + k}$. These yield the following problem-specific optimality conditions. A brief derivation is provided in Supplementary Materials A.1.

$$CR_i\Big(1 - (1 - y_i)p_i\Big) = \sum_{u \in U} P_u\Big(1 - (1 - y_i)b_{iu}\Big)F(Q_u) \qquad \text{for } Q_i > 0 \tag{6}$$

$$CR_i\Big(1 - (1 - y_i)p_i\Big) \leq \sum_{u \in U} P_u\Big(1 - (1 - y_i)b_{iu}\Big)F(Q_u) \qquad \text{for } Q_i = 0 \tag{7}$$

For the remainder of this section, we will refer to these first-order conditions as the optimality conditions.

4.1 Cost-based supplier preference order

This section characterizes the supplier preference order in the risk-neutral newsvendor model with unreliable suppliers. We show that suppliers are prioritized solely by unit procurement cost; disruption risk cannot overturn this cost-based preference order. At the same time, the disruption probability p_i and the fractional delivery guar-

[1] We refer to a disruption scenario simply as a scenario throughout this paper.

antee y_i shape the optimal order quantities and service levels, and thereby influence the activation thresholds and order allocation within this cost-based order. All proofs are given in Supplementary Materials A.

Proposition 4.1 *Let $y_k = 0$ for all suppliers. Consider two suppliers i and j ($i \neq j$). If $c_i < c_j$ and $Q_i = 0$ in an optimal solution, then it follows that $Q_j = 0$ as well.*

Proposition 4.1 establishes that if a supplier with a lower procurement cost is excluded in the optimal solution, then no supplier with a higher procurement cost will be included. Consequently, suppliers should be prioritized in non-decreasing order of their procurement costs when determining the optimal set of suppliers to order from. Proposition 4.1 is inline with Mohammadivojdan et al. (2022) for the AON setting. We next generalize this proposition including suppliers with FDGs.

Proposition 4.2 *Let $0 \leq y_k < 1$ for all suppliers. Consider two suppliers i and j ($i \neq j$). If $c_i < c_j$ and $Q_i = 0$ in an optimal solution, then it follows that $Q_j = 0$ as well.*

Proposition 4.2 should be read as an *order-qualifier* statement: unit procurement cost determines the *preference order* in which suppliers can enter an optimal portfolio. That is, if a lower-cost supplier is not used, then no higher-cost supplier can be used either. However, as we show in Sect. 4.2, the *activation* of the next supplier in this cost order, and the resulting *allocation* across active suppliers, are governed by the reliability parameters of the currently active suppliers.

For the FDG setting, Proposition 4.2 contrasts with the findings of Ray and Jenamani (2016b), who claim that suppliers should be prioritized based on their *effective cost*, defined as $c_i^{\text{eff}} = c_i E(R_i) = c_i[p_i y_i + (1 - p_i)]$. It is interesting to note that c_i^{eff} is decreasing in p_i (disruption probability) and increasing in y_i (yield under risk), which means that suppliers with higher disruption risk and lower yield under risk are favored in Ray and Jenamani (2016b). However, Proposition 4.2 shows that disruption probability and yield under risk do not alter the *cost-based preference order* of suppliers. Furthermore, Ray and Jenamani (2016b) propose a procedure that repeatedly solves the problem using a commercial nonlinear solver for the top k suppliers ($k = 1, \ldots, n$), starting with the supplier with the lowest *effective cost* in a sorted list. However, since suppliers are ordered by *effective cost* rather than unit cost, this procedure does not guarantee optimality. For a numerical example demonstrating how ordering suppliers by *effective cost* can lead to suboptimal solutions when applying the algorithm in Ray and Jenamani (2016b), see Supplementary Materials B.1.

4.2 Single-sourcing analysis and activation mechanism

To better understand how the cost-based preference order translates into actual supplier activation, we next study the single-sourcing regime in which only the lowest-cost supplier is active. This setting makes it possible to characterize the resulting order quantity and service level under a general demand distribution, and to show how these determine whether the next supplier in the cost order remains inactive or

becomes active. We then specialize the analysis to Uniform demand, which yields further closed-form characterizations and sharper comparative statistics. Throughout this section, suppliers are indexed in strictly increasing unit procurement cost[2]: $(c_1 < c_2)$.

4.2.1 Single sourcing

We begin with the single-sourcing regime in which only Supplier 1, the lowest-cost supplier, is active. In this case, the general optimality conditions reduce to a single critical-ratio equation that characterizes the optimal order quantity. This benchmark also provides the basis for studying when the next supplier in the cost order remains inactive or becomes active.

Corollary 4.3 *In a single-sourcing setting where only Supplier 1 is active, the optimal order quantity Q_1^* satisfies the following condition: $CR_1 = \frac{(1-p_1)F(Q_1^*)+y_1 p_1 F(y_1 Q_1^*)}{1-p_1+y_1 p_1}$.*

An immediate implication of Corollary 4.3 is that the optimal order quantity is at least as large as the classical newsvendor quantity. Let $q_1^{\mathrm{NV}} := F^{-1}(CR_1)$ denote the classical newsvendor benchmark; then $Q_1^* \geq q_1^{\mathrm{NV}}$.

Corollary 4.4 *In the single-sourcing setting where Supplier 1 is active, any additional supplier (e.g., Supplier 2) with a higher cost c_2 will remain inactive if the following condition holds: $CR_2 \leq (1-p_1)F(Q_1^*) + p_1 F(y_1 Q_1^*)$.*

Corollary 4.4 shows that Supplier 2 remains inactive as long as the expected service level achievable under single sourcing from Supplier 1, $SL_1 := (1-p_1)F(Q_1^*) + p_1 F(y_1 Q_1^*)$, is at least as large as Supplier 2's critical ratio, i.e., as long as $\mathrm{SL}_1 \geq \mathrm{CR}_2$. Equivalently, activating Supplier 2 requires a sufficiently low unit cost c_2 relative to the attainable service level under Supplier 1. In particular, Supplier 2's own reliability parameters (lowering p_2 or increasing y_2) do not affect this activation condition. By contrast, Supplier 1's reliability parameters affect the attainable service level under single sourcing (SL_1) and can therefore determine whether Supplier 2 becomes active. Supplementary Materials B.2 provides a further numerical illustration of these activation and allocation effects.

We next examine how disruption probability p_1 affects the optimal order quantity in the single-sourcing setting.

Corollary 4.5 *In the single-sourcing setting where Supplier 1 is active, assume $0 < y_1 < 1$ and let $Q_1^*(p_1)$ denote the optimal order quantity as a function of the disruption probability $p_1 \in [0,1]$. Then the optimal order quantity $Q_1^*(p_1)$ is increasing in the disruption probability p_1, and it satisfies $q_1^{\mathrm{NV}} \leq Q_1^*(p_1) \leq \frac{q_1^{\mathrm{NV}}}{y_1}$, $\forall\, p_1 \in [0,1]$, with $Q_1^*(0) = q_1^{\mathrm{NV}}$ and $Q_1^*(1) = q_1^{\mathrm{NV}}/y_1$.*

[2] This strict ordering is imposed only for notational simplicity. If some suppliers have equal unit costs, the results continue to hold under an arbitrary tie-breaking rule.

Corollary 4.5 shows that, under a fractional delivery guarantee with $0 < y_1 < 1$, a higher disruption probability p_1 requires a larger order quantity. In particular, the optimal order quantity $Q_1^*(p_1)$ is increasing in $p_1 \in [0, 1]$, with $Q_1^*(0) = q_1^{\mathrm{NV}}$ and $Q_1^*(1) = q_1^{\mathrm{NV}}/y_1$.

We next turn from disruption probability to the fractional delivery guarantee. Holding p_1 fixed, we study how y_1 affects both the optimal order quantity and the resulting service level.

Corollary 4.6 *In the single-sourcing setting where Supplier 1 is active, assume $0 < p_1 < 1$ and let $Q_1^*(y_1)$ denote the optimal order quantity as a function of the fractional delivery guarantee $y_1 \in (0, 1)$. Define*

$$G(z) := F(z) + zf(z), \qquad z > 0.$$

Suppose there exists a unique $\bar{z} \in (0, \infty)$ such that

$$G(z) < CR_1 \quad for z < \bar{z}, \qquad G(\bar{z}) = CR_1, \qquad G(z) > CR_1 \quad for z > \bar{z}.$$

Then $Q_1^*(y_1)$ is unimodal on (0, 1): there exists a unique $\bar{y}_1 \in (0, 1)$ such that $Q_1^*(y_1)$ is increasing on $(0, \bar{y}_1)$ and decreasing on $(\bar{y}_1, 1)$, with turning point $\bar{y}_1$ $Q_1^*(\bar{y}_1) = \bar{z}$.

Corollary 4.6 shows that the unimodality of $Q_1^*(y_1)$ follows from a single-crossing property of the function $G(z) = F(z) + zf(z)$. This condition is satisfied by several demand distributions, including Uniform$(0, b)$, Gamma(k, θ) with $k \geq 1$, Weibull(k, λ) with $k \geq 1$, and Lognormal(m, s); for Normal demand, it holds on the nonnegative domain $z \geq 0$. The corresponding verifications are provided in the Supplementary Materials B.3. The Supplementary Materials B.4, also reports similar numerical patterns for additional continuous demand distributions, including Logistic, Laplace, and Triangular. We next study how the fractional delivery guarantee affects the expected service level.

Corollary 4.7 *In the single-sourcing setting where Supplier 1 is active, assume $0 < p_1 < 1$ and let $Q_1^*(y_1)$ denote the optimal order quantity as a function of the fractional delivery guarantee $y_1 \in [0, 1]$. Then $SL_1(y_1)$ is increasing on [0, 1], with $SL_1(0) = (1 - p_1)CR_1$, and $SL_1(1) = CR_1$.*

According to Corollary 4.7, for a fixed disruption risk $p_1 \in (0, 1)$, improving the FDG y_1 (i.e., receiving a larger fraction of the order when a disruption occurs) increases the expected service level. In particular, the service level starts at $SL_1(0) = (1 - p_1)CR_1$ when disruptions wipe out deliveries ($y_1 = 0$) and increases up to $SL_1(1) = CR_1$ when delivery is unaffected ($y_1 = 1$).

4.2.2 Uniformly distributed demand

In this section, we specialize the single-sourcing analysis to the case in which demand is uniformly distributed on $(0, b)$ with $b > 0$. We first characterize the optimal order

quantity and the corresponding expected service level when Supplier 1 is the sole source. We then examine how the expected service level varies with the disruption probability p_1, how the optimal order quantity varies with the fractional delivery guarantee y_1, and finally derive a condition under which Supplier 1 remains the only active supplier.

Proposition 4.8 *Let $D \sim \text{Unif}(0, b)$. When Supplier 1 is selected as the single source, the newsvendor's optimal order quantity is given by:*

$$Q_1^* = \begin{cases} b\left(\dfrac{CR_1(1-p_1+y_1p_1)}{1-p_1+y_1^2p_1}\right), & CR_1 \le \dfrac{1-p_1+y_1^2p_1}{1-p_1+y_1p_1}, \\ b\left(\dfrac{CR_1(1-p_1+y_1p_1)-(1-p_1)}{y_1^2p_1}\right), & CR_1 > \dfrac{1-p_1+y_1^2p_1}{1-p_1+y_1p_1}. \end{cases}$$

Note that when $y_1 = 0$ (AON supplier) or $y_1 = 1$ (fully reliable supplier), the optimal order quantity is given by $Q_1^* = bCR_1$, which is the same as the order quantity of a classical newsvendor.

Using the optimal order quantity from Proposition 4.8, the corresponding expected service level $SL_1 := p_1F(y_1Q_1^*) + (1-p_1)F(Q_1^*)$ is given in Corollary 4.9.

Corollary 4.9 *Let $D \sim \text{Unif}(0, b)$. When Supplier 1 is selected as the single source, the expected service level is given by:*

$$SL_1 = \begin{cases} \frac{(1-p_1+y_1p_1)^2}{1-p_1+y_1^2p_1}CR_1 & if\ CR_1 \le \frac{1-p_1+y_1^2p_1}{1-p_1+y_1p_1} \\ (1-p_1) + \left(\frac{CR_1(1-p_1+y_1p_1)-(1-p_1)}{y_1}\right) & otherwise \end{cases} \tag{8}$$

Note that, if $y_1 = 0$, then $SL_1 = (1-p_1)CR_1$, and if $y_1 = 1$, then $SL_1 = CR_1$.

The next proposition characterizes how the expected service level varies with the disruption probability under Uniform demand.

Proposition 4.10 *Let $D \sim \text{Unif}(0, b)$. When Supplier 1 is selected as the single source, the expected service level $SL_1(p_1)$ satisfies:*

(a) *If $y_1 = 0$, then $SL_1(p_1) = (1-p_1)CR_1$ and $SL_1(p_1)$ decreases in p_1.*
(b) *If $0 < y_1 < 1$, then $SL_1(p_1)$ is unimodal in p_1: it initially decreases and then increases.*
(c) *If $y_1 = 1$, then $SL_1(p_1) = CR_1$ and $SL_1(p_1)$ is independent of p_1.*

When $y_1 = 0$, no items are delivered to the retailer under disruption (AON setting). As the disruption probability p_1 increases, the retailer's likelihood of receiving no items increases, leading to a lower service level. When $y_1 = 1$, there is no risk as the entire order is always delivered, making the service independent of p_1. The FDG case where $0 < y_1 < 1$ is less straightforward. As p_1 grows, the uncertainty in getting the full order quantity increases, leading to a lower service level. However, once p_1 surpasses a certain threshold, the retailer becomes more certain that only a portion of the order will be delivered, which allows her to adjust the order quantity accordingly,

resulting in a higher service level. Thus, the service level is unimodal in p_1, decreasing initially and then increasing after a certain threshold.

This mechanism is not unique to the Uniform case. More generally, substituting the single-sourcing critical-ratio condition (Corollary 4.3) into the service-level expression yields

$$SL_1(p_1) = CR_1 - p_1(1 - y_1)\Delta(p_1), \tag{9}$$

where $\Delta(p_1) := CR_1 - F\big(y_1 Q_1^*(p_1)\big) \geq 0$. Hence $SL_1(p_1) \leq CR_1$, and the shortfall below CR_1 is governed by the product $p_1(1 - y_1)\Delta(p_1)$. A higher p_1 raises exposure to the disrupted state, but it also raises the optimal order quantity $Q_1^*(p_1)$, which increases $y_1 Q_1^*(p_1)$ and shrinks $\Delta(p_1)$. We also observe similar numerical patterns for several additional continuous demand distributions, including Normal, Gamma, Weibull, Lognormal, Logistic, Laplace, and Triangular; see Supplementary Materials B.4.

We next characterize the conditions under which the newsvendor sources from a single supplier.

Proposition 4.11 *Let $D \sim \mathrm{Unif}(0, b)$. The newsvendor orders only from Supplier 1 if the following condition holds:*

$$y_1 \geq \begin{cases} 2\frac{CR_1}{CR_2} - 1 - 2\sqrt{\frac{CR_1}{CR_2}\left(\frac{CR_1}{CR_2} - 1\right)} & if\ CR_2 > \frac{2CR_1 - 1}{CR_1} \\ CR_2 \frac{1 - CR_1}{1 - CR_2} & if\ CR_2 \leq \frac{2CR_1 - 1}{CR_1} \end{cases}$$

The condition in Proposition 4.11 can be interpreted as a threshold on the lowest-cost supplier's yield guarantee under disruption such that the lowest-cost supplier is selected as the single source if its yield under risk is high enough. This threshold decreases in the ratio $\frac{CR_1}{CR_2} = \frac{s - c_1 + k}{s - c_2 + k}$. Hence, the newsvendor expects a better performance from the lowest-cost supplier as the second supplier becomes more competitive in terms of procurement cost. In particular, the threshold remains unaffected by the probability of disruption associated with the lowest-cost supplier.

After presenting analytical results for the Uniform distribution, we turn to a Normal-demand numerical example and observe similar patterns in the next section.

4.3 Numerical example

We study how a supplier's disruption probability p_i and yield under disruption y_i affect the retailer's optimal order quantities and the resulting expected service level. We use the same dataset as Ray and Jenamani (2016b): demand is Normal with mean 400 and standard deviation 130, ten suppliers are available, and we set $k = 15$, $s = 45$, and $r = 10$. In the baseline optimal portfolio, Suppliers 2 and 5 are active; Table 1 reports their parameters.

We proceed in two steps. First, we analyze a *single-sourcing* benchmark in which the retailer orders exclusively from Supplier 5 to isolate the mechanisms driving the

order-quantity and service-level responses. We then return to the *two-supplier* setting (Suppliers 2 and 5) to quantify how the activation of Supplier 2 interacts with Supplier 5's disruption risk and yield.

4.3.1 Single-sourcing benchmark (Supplier 5 only)

We first consider a single-sourcing benchmark in which the retailer orders exclusively from Supplier 5. Since Supplier 2 is not available in this benchmark, the optimal order quantity Q_5^* satisfies the critical-ratio condition

$$CR_5 = \frac{(1-p_5)F(Q_5^*) + y_5 p_5 F(y_5 Q_5^*)}{1 - p_5 + y_5 p_5} = 0.78, \tag{10}$$

and the corresponding expected service level is

$$SL_5 = (1-p_5)F(Q_5^*) + p_5 F(y_5 Q_5^*). \tag{11}$$

Figure 1 reports the Normal-demand benchmark along two dimensions: panels (a) and (b) vary the disruption probability p_5 with $y_5 = 0.4$ fixed, whereas panels (c) and (d) vary the FDG level y_5 with $p_5 = 0.2$ fixed.[3]

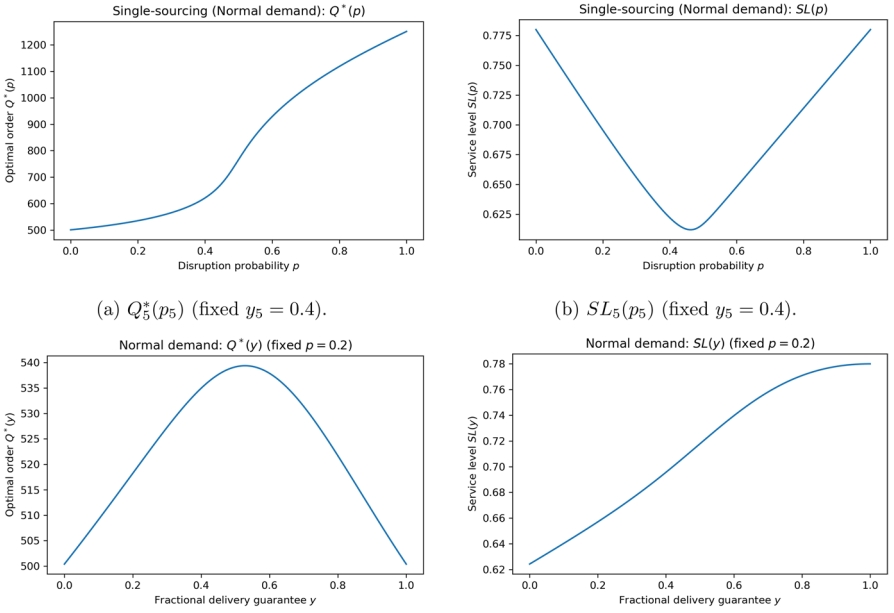
Figure 1a shows that $Q_5^*(p_5)$ increases with p_5, consistent with Corollary 4.5. The service level $SL_5(p_5)$, defined by (11), appears to be non-monotone in p_5, following a decrease–then–increase pattern (Fig. 1b). This pattern is established analytically for Uniform demand in Proposition 4.10; here we observe the same behavior under Normal demand. Moreover, (11) implies that $SL_5(0) = SL_5(1) = CR_5 = 0.78$.

Figure 1c and d show the effects of the fractional delivery guarantee y_5. We observe that $SL_5(y_5)$ increases in y_5, from $SL_5(0) = (1-p_5)CR_5 = (1-0.2) \times 0.78 = 0.624$ to $CR_5 = 0.78$ as $y_5 \to 1$. This is consistent with Corollary 4.7: a higher fractional delivery guarantee improves the delivered quantity in the disruption state and therefore improves service. The optimal order quantity is non-monotone in y_5. Since the condition in Corollary 4.6 is satisfied for Normal demand on the nonnegative demand region, $Q_5^*(y_5)$ is unimodal: it increases for small y_5, attains an interior maximum, and then decreases as $y_5 \to 1$. The turning point can be computed numerically by first solving $F(\bar{z}) + \bar{z} f(\bar{z}) = CR_5$, which yields $\bar{z} \approx 285.05$, and then solving $y Q_5^*(y) = \bar{z}$, which yields $\bar{y} \approx 0.5285$ and $Q_5^*(\bar{y}) \approx 539.38$. Moreover, $Q_5^*(y_5)$ coincides with the classical newsvendor quantity $q_5^{\mathrm{NV}} := F^{-1}(CR_5)$ at the endpoints $y_5 \in \{0, 1\}$.

Table 1 Supplier data for Supplier 2 and Supplier 5

Supplier	p_i	y_i (%)	c_i	c_i^{eff}	CR_i	$E(R_i)$	Var(R_i)
2	0.25	40	25	21.3	0.70	0.85	0.07
5	0.20	40	21	18.5	0.78	0.88	0.06

[3] We use (E) to denote risk-neutral results.



(a) $Q_5^*(p_5)$ (fixed $y_5 = 0.4$).

(b) $SL_5(p_5)$ (fixed $y_5 = 0.4$).

(c) $Q_5^*(y_5)$ (fixed $p_5 = 0.2$).

(d) $SL_5(y_5)$ (fixed $p_5 = 0.2$).

Fig. 1 (E) Single-sourcing (Supplier 5 only): Q_5^* and SL_5 under Normal demand. Top: vary p_5 (fixed $y_5 = 0.4$). Bottom: vary y_5 (fixed $p_5 = 0.2$)

4.3.2 Two-supplier case (Suppliers 2 and 5)

We now return to the two-supplier setting and allow the retailer to order from Supplier 2 in addition to Supplier 5. Building on the single-sourcing benchmark, the key question is when the backup supplier becomes active and how the allocation (Q_2, Q_5) adjusts as Supplier 5's reliability parameters change.

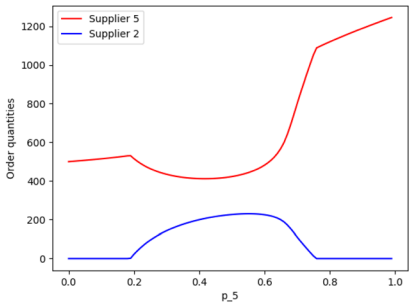
We proceed in three steps. First, we vary Supplier 5's disruption probability p_5 for a fixed FDG level $y_5 = 0.4$ and identify the range of disruption probabilities in which Supplier 2 is activated. Second, we vary the FDG level y_5 for a fixed disruption probability $p_5 = 0.2$ to quantify how improved yield under risk reduces reliance on Supplier 2. Finally, we consider the joint effect of (p_5, y_5) by plotting order quantities and service levels as functions of p_5 for multiple FDG levels y_5.

We vary Supplier 5's disruption probability p_5 while keeping all other parameters fixed and compare two cases for Supplier 5's yield under disruption: an FDG case with $y_5 = 0.4$ (Fig. 2) and an all-or-nothing (AON) case with $y_5 = 0$ (Fig. 3).

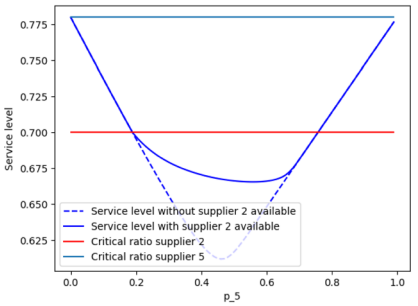
Supplier 2 remains inactive as long as the service level attainable under single sourcing from Supplier 5 stays above Supplier 2's critical ratio:

$$CR_2 \le SL_5 := (1 - p_5)F(Q_5) + p_5 F(y_5 Q_5), \qquad (CR_2 = 0.70). \tag{12}$$

Since $SL_5 = CR_5$ at $p_5 \in \{0, 1\}$, Supplier 2 is inactive at both endpoints. For intermediate disruption risks, however, SL_5 can fall below CR_2, which activates Sup-

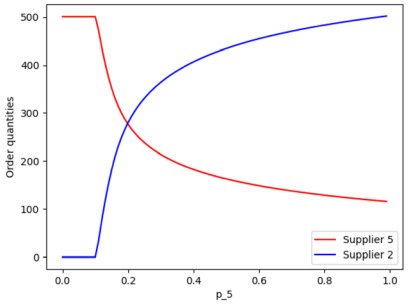


(a) Order quantities of suppliers 2 and 5.

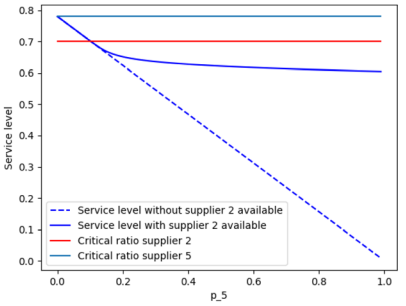


(b) Service level.

Fig. 2 (E) Two-supplier case: order quantities and service level versus p_5 when Supplier 5 provides a fractional delivery guarantee (FDG) with $y_5 = 0.4$



(a) Order quantities of suppliers 2 and 5 as functions of p_5.



(b) Service level as a function of p_5.

Fig. 3 (E) Two-supplier case: order quantities and service level versus p_5 when Supplier 5 is under the all-or-nothing (AON) setting ($y_5 = 0$)

plier 2; in our numerical example this occurs for $0.19 < p_5 < 0.75$. Thus, Supplier 5 remains the sole source for $p_5 < 0.19$ and $p_5 > 0.75$, whereas both suppliers are used in the intermediate range.

The AON case highlights why fractional delivery guarantee matters. When $y_5 = 0$, Supplier 5 cannot deliver in the disruption state and the optimality conditions reduce to $CR_5 = F(Q_5)$ and $CR_2 \leq (1 - p_5)F(Q_5)$. Hence Supplier 5 is selected exclusively when $p_5 \leq 1 - \frac{CR_2}{CR_5}$, in which case Q_5 equals the classical newsvendor quantity and does not vary with p_5 (Fig. 3a). The corresponding single-sourcing service level, $(1 - p_5)F(Q_5)$, decreases in p_5; once it falls below CR_2, Supplier 2 becomes active (Fig. 3b). After activation, the allocation shifts away from Supplier 5 and its order quantity decreases (but remains nonnegative).

Next, we examine the impact of FDG by varying y_5 while keeping all other parameters fixed (in particular, $p_5 = 0.2$). In Fig. 4b, for low y_5 the single-sourcing service level (dashed curve) lies below CR_2, so Supplier 2 is active and the realized service

level with Supplier 2 available (solid curve) is higher. As y_5 increases, the single-sourcing service level rises and crosses the threshold CR_2 at approximately $y_5 \approx 0.4$, after which Supplier 2 becomes inactive and the two service-level curves coincide. Finally, as $y_5 \to 1$, Supplier 5 becomes effectively reliable and the service level converges to $CR_5 = 0.78$.

Figure 4a shows the corresponding order quantities. As y_5 increases, Q_2 decreases monotonically and drops to zero once y_5 is sufficiently high. Over the region in which Supplier 2 is active, Q_5 increases with y_5. Once single sourcing resumes, Q_5 follows the unimodal pattern characterized in Corollary 4.6: it continues to increase slightly up to the turning point $\bar{y} \approx 0.5285$, where $Q_5^*(\bar{y}) \approx 539.38$, and then decreases mildly as $y_5 \to 1$, converging to the classical newsvendor quantity $q_5^{\mathrm{NV}} := F^{-1}(CR_5) \approx 500.39$.

We next study the joint effect of Supplier 5's disruption probability p_5 and yield under risk (FDG level) y_5, while keeping Supplier 2's parameters fixed. Figure 5 reports the resulting order quantities and expected service levels as functions of p_5 for several values of y_5.

Figure 5 shows a clear substitution pattern as Supplier 5 becomes more resilient. When $y_5 = 1$, Supplier 5 is perfectly reliable, Supplier 2 is not utilized, and the expected service level equals CR_5 for all p_5. For high but imperfect yields (e.g., $y_5 = 0.8$ and $y_5 = 0.6$), the retailer continues to source exclusively from Supplier 5 across the entire p_5 range, consistent with Proposition 4.11; in this region the service level remains unimodal in p_5 and stays above $CR_2 = 0.70$. For lower yields (e.g., $y_5 = 0.4$ and $y_5 = 0.2$), there exists an intermediate range $p_5^l < p_5 < p_5^h$ in which the implied service level from sourcing only from Supplier 5 falls below CR_2, activating Supplier 2. As y_5 increases, this activation range narrows and the order quantity allocated to Supplier 2 decreases. Across all FDG levels with $0 < y_5 < 1$, the expected service level remains unimodal in p_5, in line with the analytical unimodality result established for Uniform demand (Proposition 4.10).

A key contrast with the AON case ($y_5 = 0$; Mohammadivojdan et al. 2022) lies in the service-level response: under AON, the single-sourcing service level decreases in p_5, whereas for $0 < y_5 < 1$ it is unimodal, decreasing first and then increasing in p_5. This difference also affects supplier activation. In the AON case, increasing the disruption probability of the lowest-cost supplier does not lead to the re-elimination

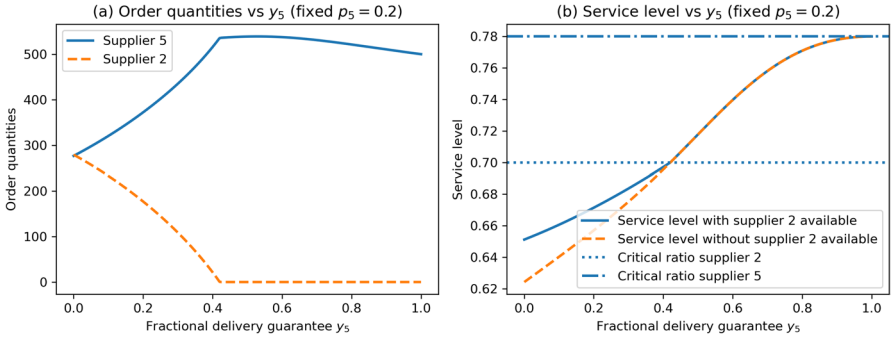
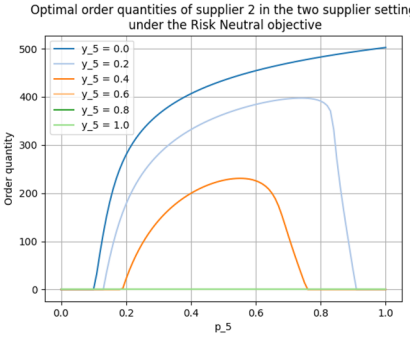
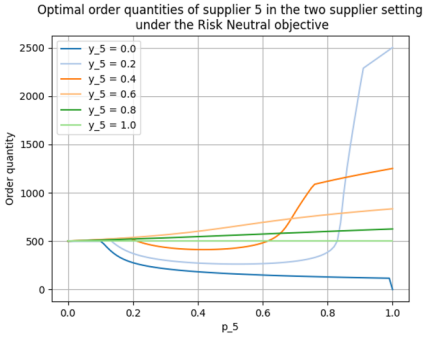


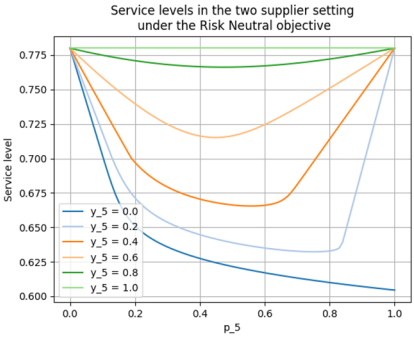
Fig. 4 (E) Two-supplier case: order quantities and service level versus y_5 with $p_5 = 0.2$ fixed



(a) Order quantity Q_2 for various values of y_5.



(b) Order quantity Q_5 for various values of y_5.



(c) Service level for various values of y_5.

Fig. 5 (E) Two-supplier case: order quantities and service level as functions of p_5 for different fractional delivery guarantee (FDG) levels y_5

of a higher-cost supplier once that supplier becomes active. By contrast, such re-elimination can occur in the fractional-yield setting.

5 Risk-averse newsvendor with unreliable suppliers

This section focuses on a risk-averse newsvendor with multiple unreliable suppliers. We utilize variance (V) and weighted variance (WV) as the risk measures and consider the mean-risk approach. Section 5.1 introduces the variance and weighted variance functions, and compares them in a single supplier setting. Next, in Sect. 5.2, we analyze mean-(weighted) variance problem. We then introduce our solution approach for the the mean-(weighted) variance problem in Sect. 5.3.

5.1 Variance and weighted variance

For a given disruption scenario u, the variance of the profit can be expressed as

$$G_V(Q_u) = V[G(Q_u; D)] = (s-r)^2 n_2(Q_u) + k^2 \ell_2(Q_u) - \left[(s-r)n(Q_u) + k\ell(Q_u)\right]^2. \tag{13}$$

Based on the *law of total variance*, the variance can be expressed as

$$G_V(Q) = E[G_V(Q_u)] + Var[G_E(Q_u)] \tag{14}$$

where $E[G_V(Q_u)] = \sum_{u\in U} P_u G_V(Q_u)$, $Var[G_E(Q_u)] = \sum_{u\in U} P_u\ [G_E\ (Q_u)]^2 - [\sum_{u\in U} P_u\, G_E\ (Q_u)]^2$.

Note that in Ray and Jenamani (2016a), the variance of the profit, where suppliers are subject to risk, only includes the weighted average of the variances over all risk scenarios, i.e., $G_{WV}(Q) = E[G_V(Q_u)]$, disregarding $\mathrm{Var}[G_E(Q_u)]$. While the variance is only partially captured with this objective, *the weighted variance* (Eq. 15) can still be used as a risk measure due to its simplicity.

$$G_{WV}(Q) = \sum_{u\in U} P_u G_V(Q_u) = \sum_{u\in U} P_u \Big\{(s-r)^2 n_2(Q_u) + k^2 \ell_2(Q_u) - \left[(s-r)n(Q_u) + k\ell(Q_u)\right]^2\Big\}. \tag{15}$$

Next, we present a proposition that suggests that a risk-averse newsvendor utilizing weighted variance objective prefers to diversify the supplier portfolio by ordering from all available suppliers.

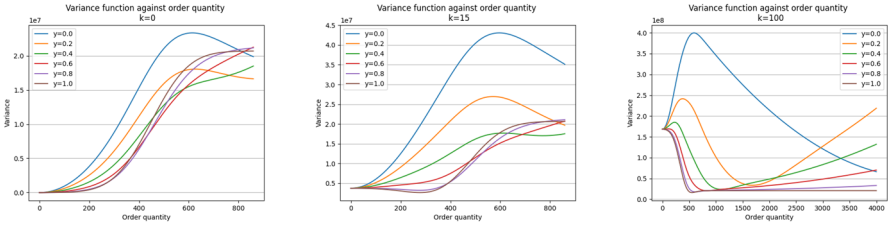
Proposition 5.1 *Let $0 \le y_i < 1$ for every supplier i and $k > 0$. Let demand follow a symmetric distribution with IHR or a distribution with IHR and DRHR. Then, a newsvendor minimizing the weighted variance (WV) will always order some amount from every available supplier, i.e., no supplier will be inactive.*

Further analysis of the variance and weighted variance can be found in Supplementary Materials B.5. Due to its extreme complexity, we further characterize the properties of the variance and weighted functions by considering the single sourcing setting.

5.1.1 Special case: single AON supplier

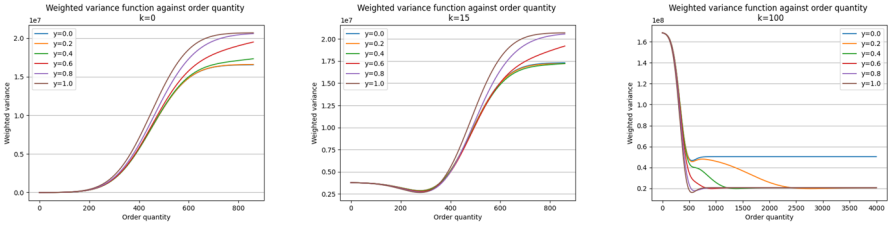
We consider the single supplier setting, as discussed in Supplementary Materials B.6. Noting that $G(Q; D)$ denotes the newsvendor's profit function (with a single, reliable supplier), we find that:

$$\begin{aligned} G_E(Q) &= (1-p)E[G(Q;D)] + pE[G(yQ;D)], \\ G_{WV}(Q) &= (1-p)V[G(Q;D)] + pV[G(yQ;D)], \\ G_V(Q) &= (1-p)(E[G(Q;D) - G_E(Q)])^2 \\ &\quad + p(E[G(yQ;D) - G_E(Q)])^2 + G_{WV}(Q). \end{aligned}$$



(a) $k = 0$ (b) $k = 15$ (c) $k = 100$

Fig. 6 Variance (V) results - fractional delivery guarantee (FDG), $p = 0.2$



(a) $k = 0$ (b) $k = 15$ (c) $k = 100$

Fig. 7 Weighted Variance (WV) results - fractional delivery guarantee (FDG), $p = 0.2$

Note that for the AON supplier setting ($y = 0$), when there is no stockout cost ($k = 0$), we have

$$G_E(Q) = (1-p)E[G(Q;D)], G_{WV}(Q) = (1-p)V[G(Q;D)],$$
$$G_V(Q) = p(1-p)(E[G(Q;D)])^2 + (1-p)V[G(Q;D)].$$

As the supplier's disruption probability p increases, both the expected profit and the weighted variance decrease, by a factor $(1-p)$, compared to the case where the supplier is fully reliable. Note that when the reliability variance, $p(1-p)$, is negligible (that is, p is close to 0 or 1), $G_{WV}(Q)$ closely approximates the variance; otherwise, the difference becomes greater. Furthermore, when p is very high, the variance term becomes small because the supplier rarely delivers, leading to more predictable (though lower) outcomes. This suggests that under a risk-averse framework utilizing variance as a risk measure, a highly unreliable supplier may still be preferable in extreme cases where its limited contribution leads to lower variance in overall profit.

5.1.2 Numerical example for one supplier

We have conducted a numerical study to examine the shape of the variance function (Eq. 14) and weighted variance function (Eq. 15) in a single unreliable supplier setting. This analysis is based on the numerical setting introduced in Sect. 4.3, focusing on the lowest-cost supplier, Supplier 5, we vary y_5 and p_5. Figures 6 and 7 presents the variance (weighted variance) results under the FDG setting, with $p_5 = 0.2$ and varying values of y_5 and stockout cost k. Note that in these figures, the case $y = 0$

represents the AON setting, while $y = 1$ corresponds to the classical newsvendor setting with a reliable supplier. Table 2 provides the order quantities at which the optimal solution is observed given the objective.

When the stockout cost is zero $(k = 0)$, the weighted variance objective increases with the order quantity, regardless of the supplier's unreliability. Note that, in the AON setting, the weighted variance is $(1 - p)V[G(Q; D)]$. Since $V[G(Q; D)]$ is an increasing function, this result is expected. In contrast, in this setting, the variance increases with the quantity of orders, peaks, and then decreases for large orders. Note that this peak is closer to the peak observed in the expected profit function, $(1 - p)E[G(Q; D)]$, as for the AON setting $G_V(Q) = p(1 - p)(E[G(Q; D)])^2 + (1 - p)V[G(Q; D)]$. In the FDG setting, lowering the shipment guarantee, y_5, accentuates the peak, while as y_5 approaches 1, the variance curve flattens. When stockout costs are introduced $(k > 0)$, the difference between the weighted variance and the variance becomes more visible, affecting both the magnitude and distribution of risk. The weighted variance becomes a unimodal function with a single stationary point, first decreasing and then increasing with order quantity, regardless of the supplier's risk. On the other hand, with stockout costs, the variance function is unimodal (decreasing and then increasing, consistent with Fleuren 2025) only when the supplier is reliable. When the supplier is unreliable, the variance function may exhibit multiple stationary points.

According to Table 2, when $k = 0$, both weighted variance and variance are minimized at zero, regardless of supplier risk. We next examine the solutions for cases where $k > 0$. In the AON setting $(y = 0)$, the supplier's disruption risk (p) does not impact the order quantity for a newsvendor minimizing weighted variance. The resulting order quantity mirrors that of the classical newsvendor model, where supplier reliability is not a concern. In the FDG setting, as y increases, the order quantity initially rises before declining, ultimately converging to the classical newsvendor solution when the supplier becomes fully reliable. A variance-minimizing newsvendor adopts a different ordering strategy for AON supplier compared to a riskless supplier. When the stockout cost is relatively low $(k = 15)$, the newsvendor tends to avoid ordering from the AON supplier. However, the order quantity increases as the yield guarantee (y) improves. This behavior reverses when the stockout cost is extremely high $(k = 100)$, leading the newsvendor to order more from the supplier providing lower guarantee. Note that in this setting, the order quantity that minimizes risk may increase to the level that maximizes expected profit.

5.2 Mean-(weighted) variance problem

The Mean–variance objective for the newsvendor can be stated as

$$G_{MV(\alpha)}(Q) = G_E(Q) - \alpha G_V(Q) \tag{16}$$

where $G_E(Q)$ and $G_V(Q)$ represent expectation and variance of profit, respectively. For an active supplier $(Q_i > 0)$, the first-order condition can be written as:

Table 2 Comparison of optimal order quantities maximizing E, minimizing V and WV objectives for AON ($y = 0$) and FDG ($p = 0.2$) cases

AON ($y = 0$)							FDG ($p = 0.2$)						
	$k \setminus p$	0	0.2	0.4	0.6	0.8	$k \setminus y$	0	0.2	0.4	0.6	0.8	1
E	0	463	463	463	463	463	0	463	476	487	491	481	463
	15	500	500	500	500	500	15	500	518	535	538	522	500
	100	581	581	581	581	581	100	581	633	711	669	614	581
WV	0	0	0	0	0	0	0	0	0	0	0	0	0
	15	277	277	277	277	277	15	277	278	279	281	283	277
	100	553	553	553	553	553	100	553	2766	1383	921	603	553
V	0	0	0	0	0	0	0	0	0	0	0	0	0
	15	277	1	0	0	0	15	1	1	3	7	275	277
	100	553	3999	3999	3999	3999	100	3999	1682	1106	837	614	553

$$\frac{(s-c_i+k)[1-(1-y_i)p_i]}{(s-r+k)} = \sum_{u\in U} P_u\Big(1-(1-y_i)b_{iu}\Big)\Big(\Delta(Q_u) + 2\alpha\big(CR_i - F(Q_u)\big)\big(G_E(Q_u) - \sum_{u\in U} P_u G_E(Q_u)\big)\Big) \tag{17}$$

where $\Delta(Q) = F(Q) - 2\alpha h(Q)$ and $h(Q) = kF(Q)\,\ell(Q) - (s-r)n\,(Q)\,(1-F(Q))$. Based on the KKT conditions, we find the following condition for an inactive supplier:

$$\frac{(s-c_i+k)[1-(1-y_i)p_i]}{(s-r+k)} \le \sum_{u\in U} P_u\Big(1-(1-y_i)b_{iu}\Big)\Big(\Delta(Q_u) + 2\alpha\big(CR_i - F(Q_u)\big)\big(G_E(Q_u) - \sum_{u\in U} P_u G_E(Q_u)\big)\Big) \tag{18}$$

Note that we cannot conclude convexity or concavity based on the second derivative, so the KKT conditions are not sufficient to guarantee an optimal solution.

We next introduce the mean-weighted variance problem where the objective function can be written as follows: $G_{MWV(\alpha)}(Q) = G_E(Q) - \alpha G_{WV}(Q)$. The necessary KKT conditions yield the following optimality conditions:

$$\frac{(s-c_i+k)[1-(1-y_i)p_i]}{(s-r+k)} = \sum_{u\in U} P_u\Big(1-(1-y_i)b_{iu}\Big)\Delta(Q_u) \text{ for } Q_i > 0 \tag{19}$$

$$\frac{(s-c_i+k)[1-(1-y_i)p_i]}{(s-r+k)} \le \sum_{u\in U} P_u\Big(1-(1-y_i)b_{iu}\Big)\Delta(Q_u) \text{ for } Q_i = 0. \tag{20}$$

We revisit the special case, as described in Sect. 5.1.1, where the supplier follows an AON policy $(y = 0)$ and there is no stockout cost $(k = 0)$. For this single sourcing setting, the Mean–variance problem can be formulated as follows where $G(Q; D)$ denote the newsvendor's profit when supplier is fully reliable: $G_{MV(\alpha)} = (1-p)\,[E[G(Q;D)] - \alpha V[G(Q;D)]] - \alpha p(1-p)(E[G(Q;D)])^2$ (see Supplementary Materials B.6). Similarly, the mean-weighted variance problem can be written as: $G_{MWV(\alpha)} = (1-p)\,[E[G(Q;D)] - \alpha V[G(Q;D)]]$. Note that for this special case, when the decision-maker utilizes weighted variance as a risk measure, the optimal order quantity remains the same as the newsvendor's mean-risk order quantity. However, when the decision-maker maximizes the Mean–variance objective, reliability influences the decision.

5.3 Solution approach for the mean-risk problem

The problem formulations presented in this study depend on an exponential number of disruption scenarios, 2^n. Nevertheless, in practice, n rarely exceeds 10. While such problems can be formulated and solved when n is relatively small, solving them becomes computationally challenging as n grows. Commercial nonlinear optimiza-

tion solvers are often employed to tackle these formulations. For instance, Ray and Jenamani (2016a, b) used MATLAB 8.0 (R2012b) to solve such problems. In this study, we use the properties of the normal distribution to formulate and solve these problems using the Conopt nonlinear solver. Note that nonlinear solvers may not always be accessible. For certain distributions, first- and second-order loss-complementary loss functions may lack closed-form expressions, making their computation more complex. Additionally, these solvers may not scale efficiently for larger instances. To address these challenges, we propose an alternative solution method that bypasses the need for nonlinear optimization solvers.

Our approach relies on root-finding techniques, specifically solving systems of equations derived from first-order conditions. It operates on the principle that, for a given set of active suppliers, the optimal order quantities can be determined by solving a system of k equations for k active suppliers. Detailed formulations can be found in Eq. 6 (risk-neutral), 19 (mean-weighted variance) and 17 (Mean–variance). Exhaustively evaluating all combinations quickly becomes computationally prohibitive as n increases. To determine the optimal order quantities and supplier selection, we use a preference order based on the lowest procurement cost. In our Algorithm 1, the suppliers are ordered according to increasing procurement costs and categorized into three sets: active (A), inactive (I), and remaining (R). Initially, all suppliers are in R. The most preferred supplier is moved to A, and the first-order conditions are solved to determine optimal order quantities. For the remaining suppliers in R, the inactivity conditions are checked, and those satisfying them are moved to I. This process repeats sequentially until R is empty. At each iteration, first-order conditions are solved for all active suppliers (A), requiring k equations for k active suppliers.

Algorithm 1 Preference-Order-Based Supplier Selection Algorithm

1. **Initialization:**
   - $A \leftarrow \emptyset$; $R \leftarrow$ All suppliers sorted by preference order; $I \leftarrow \emptyset$
2. **Iterative Supplier Selection: While** $R \neq \emptyset$:
   a. Move the next preferred supplier from R to A.
   b. Solve the first-order conditions for suppliers in A to determine Q_i. Update the objective function and evaluate its value.
   c. For each supplier remaining in R, evaluate the KKT inactivity conditions.
      - If the inactivity conditions are satisfied, move the supplier to I.
      - Otherwise, keep the supplier in R
3. **Final Check** Before finishing, check that the inactive conditions for all suppliers in I are satisfied under the optimal order quantities of the suppliers in A from the last iteration.
4. **Termination and Output:** When R is empty, the algorithm outputs:
   - Active suppliers (A) with optimal order quantities Q_i.
   - Inactive suppliers (I) with zero order quantities.
   - Final objective function values (e.g., expected profit and risk measure).

This algorithm is guaranteed to provide the optimal solution in a risk-neutral setting. For the variance minimization or weighted variance minimization setting ($\alpha \to \infty$), all suppliers are immediately added.

In the risk-averse setting, the algorithm performs well, as demonstrated in our numerical study. A risk-averse buyer tends to diversify their portfolio by ordering from multiple suppliers. While rejection decisions are still primarily guided by procurement costs, order quantities are influenced by risk factors. Note that we cannot prove the sufficiency of the KKT conditions in the risk-averse setting, leading to multiple feasible solutions. As a result, when evaluating Step 2.c in the algorithm, multiple order quantities may satisfy the conditions, requiring additional steps to ensure the best selection. We address this issue in our implementation, as discussed in Sect. 6.1.

6 Numerical study

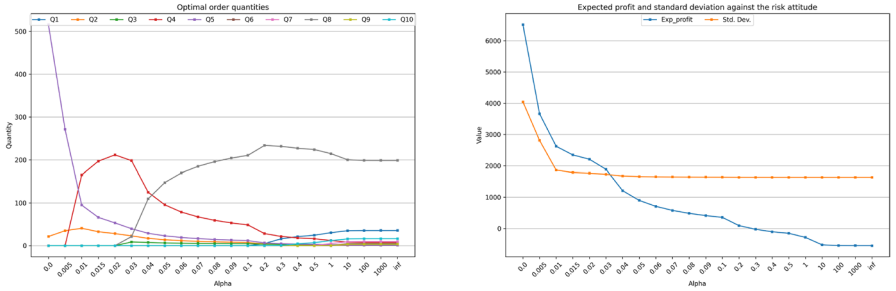
We conduct an extensive numerical study to evaluate the effectiveness of the proposed solution algorithm, assess its computational performance and compare its solutions with those obtained from a commercial solver (Sect. 6.1). We then analyze the impact of risk on firms' procurement strategies, and derive managerial insights (Sect. 6.2). Finally, Sect. 6.3 presents the sensitivity analysis on key parameters, such as stockout cost and demand variance, to explore how different levels of risk aversion influence ordering decisions. In our numerical analysis, we use the supplier data in Table 3. Other parameters are set as $k = 15$, $s = 45$, and $r = 10$, with demand modeled as a normal distribution ($\mu = 400$, $\sigma = 130$). Interestingly, ordering suppliers by procurement cost (c_i) and effective cost (c_i^{eff}) yields the same sequence for this particular instance: 5, 2, 4, 3, 8, 1, 10, 7, 9, 6.

6.1 Performance of the proposed algorithm

In this section, we assess the performance of the Algorithm 1 for varying objectives and risk levels. To benchmark our solutions, we also implemented the optimization problems in *GAMS* and solved them using the nonlinear solver Conopt (Corporation

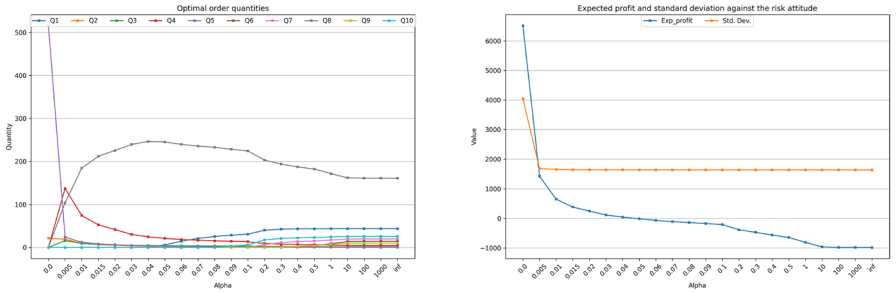
Table 3 Supplier data from Ray and Jenamani (2016a, b)

Supplier	p_i	y_i (%)	c_i	c_i^{eff}	CR_i	$E(R_i)$	Var(R_i)
1	0.05	80	40	39.6	0.40	0.99	0.00
2	0.25	40	25	21.3	0.70	0.85	0.07
3	0.15	50	35	32.4	0.50	0.93	0.03
4	0.10	70	30	29.1	0.60	0.97	0.01
5	0.20	40	21	18.5	0.78	0.88	0.06
6	0.08	60	55	53.2	0.10	0.97	0.01
7	0.12	75	45	43.7	0.30	0.97	0.01
8	0.04	90	37	36.9	0.46	1.00	0.00
9	0.17	65	52	48.9	0.16	0.94	0.02
10	0.22	85	42	40.6	0.36	0.97	0.00



(a) Order Quantities (b) Expected Profit and Risk measure

Fig. 8 MWV(α) results



(a) Order Quantities (b) Expected Profit and Risk measure

Fig. 9 MV(α) results

2024). The computational experiments were conducted on a system with an AMD Ryzen 5 3500U processor (2.10 GHz) and 16 GB of RAM. Algorithm 1 was implemented in Python using the *SciPy* package to iteratively solve the first-order conditions. In each iteration, `fsolve` is employed to solve these conditions for all active suppliers in set A, with solution precision controlled by setting the `xtol` argument to $1e-4$ and the initial guess that evenly distributes the mean demand among all suppliers in A: $Q_i = \frac{\mu}{|A|}, \quad \forall i \in A$. Using the initial guess, `fsolve` iteratively computes a solution, which is accepted if the absolute residuals of the equations are below 0.01 and all entries remain nonnegative. Based on the data in Table 3, we tested 21 risk levels (α), including the risk-neutral case ($\alpha = 0$) and the fully risk-averse case ($\alpha \to \infty$), using both MWV(α) and MV(α) objectives. In all instances except one,[4] our algorithm produced the same solution as GAMS only using the first initial guess. This problematic instance is effectively equivalent to the fully risk-averse case (see Figs. 8b and 9b).

Table 4 summarizes the running times for Algorithm 1, GAMS (including both model formulation and solver time), and Conopt (solver time only) for both the MWV(α) and MV(α) objectives (see Supplementary Materials C for complete

[4] Only for the MWV objective when $\alpha = 100$, the second initial guess is included.

Table 4 Comparison of run times (seconds) for the MWV(α) and MV(α) objectives

α	MWV objective			MV objective		
	Algorithm 1	GAMS	Conopt	Algorithm 1	GAMS	Conopt
0.000	18.63	8.18	0.45	19.21	2.14	0.27
0.010	18.03	2.56	0.36	57.10	2.37	0.31
0.1	40.63	2.42	0.38	134.69	2.50	0.47
1	361.93	2.80	0.47	1321.45	2.52	0.45
10	811.40	2.96	0.44	1370.39	2.65	0.50
100	798.61	2.55	0.47	1426.26	2.77	0.47
∞	391.27	2.62	0.53	767.90	2.70	0.53

results). The results indicate that while the commercial solver efficiently solves the problem, our algorithm experiences increasing solution times as α rises, particularly for the MV(α) objective. Despite this, our algorithm remains a viable alternative, especially for practical risk levels. Moreover, while commercial solvers effectively handle this problem, they may not always be accessible. We successfully solved this problem using Conopt in GAMS by using the properties of the normal distribution. The availability of closed-form expressions for the first- and second-order loss functions allowed us to formulate the models directly within the solver. However, for other distributions, this approach may not be feasible, as first-order conditions involve terms such as the cumulative distribution function (CDF) and loss functions, which often require integration. Unlike commercial solvers, *fsolve* can handle these integrals, ensuring that our algorithm remains effective across different distributions. Our algorithm exploits the problem's optimality conditions, eliminating the need for a commercial nonlinear solver. In contrast, Ray and Jenamani (2016b) and its extension Ray and Jenamani (2016a) rely on a nonlinear solver (MATLAB 8.0 (R2012b)) to iteratively optimize the growing subsets of suppliers. Associated computational cost scales with the subset size, ultimately requiring a full problem solution in the worst case, diminishing any advantage for large-scale instances. A direct comparison with Ray and Jenamani (2016a) is impractical due to the differing computational setups and solvers. However, their reported solution times range from 105.6 s for a single supplier to 120,000 s for six suppliers in one iteration.

6.2 Effect of risk aversion on the procurement strategy

We provide a comparative analysis of solutions under different objectives: risk-neutral (E) and risk-averse settings (mean weighted variance (MWV(α)) and mean–variance (MV(α)). We introduce risk aversion by increasing α from 0 to 1000 and solve the problem under two risk-averse objectives. Finally, we analyze scenarios where the objectives are exclusively to minimize the weighted variance (WV) and to minimize variance (V). These scenarios can be considered as $\alpha = \infty$, where risk aversion dominates the solution.

Table 5 summarizes the order quantities as well as expected profit and standard deviation for problems E, V and WV. Note that supplier columns are ordered with increasing unit procurement cost: 5, 2, 4, 3, 8, 1, 10, 7, 9, 6. In the risk-neutral setting, the solution only involves two active suppliers: Supplier 5, with the lowest cost,

Table 5 Optimal solution for the risk-neutral and risk-averse settings

Setting	Exp. Profit	St Dev	5	2	4	3	8	1	10	7	9	6
E	6516.3	4044.9	516.1	21.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
V	−980.3	1634.0	0.3	0.3	4.4	1.6	161.2	43.7	25.7	19.7	10.0	14.1
WV	−550.0	1634.2	1.0	0.8	7.8	1.9	198.4	35.5	16.3	9.6	3.5	5.4

receives $Q_5 = 516$, and Supplier 2, with the second-lowest cost, receives $Q_2 = 21$. In both risk-averse settings (V and WV), all suppliers were activated, consistent with Proposition 5.1.

Interestingly, the difference in standard deviations between the variance-minimizing (V) and weighted variance-minimizing (WV) models is negligible, indicating that $Var[G_E(Q_u)]$ (variance of expected profit among disruption scenarios) is minimal. This can be explained by ordering patterns and supplier data: both models allocate most orders to *high-reliability suppliers* with strong expected performance and low variability. For this problem setting, minimizing variance does not provide the potential benefit of reducing risk, while it significantly lowers expected profit from -550 to -980. However, this conclusion cannot be generalized to settings where delivery guarantees are lower, especially for AON settings. Expected reliability, $E(R_i) = 1 - (1 - y_i)p_i$, worsens with low y_i and high p_i. The reliability variance is given by $\mathrm{Var}(R_i) = (1 - y_i)^2 p_i(1 - p_i)$. As y_i approaches 1, $\mathrm{Var}(R_i)$ approaches to zero, eliminating uncertainty in supplier performance. Conversely, when $y_i = 0$ (AON setting), variance depends entirely on $p_i(1 - p_i)$, the intrinsic variability of a Bernoulli random variable. This variance is maximized at $p_i = 0.5$, and decreases as p_i approaches 0 or 1, representing deterministic outcomes. For intermediate values of y_i, $\mathrm{Var}(R_i)$ is scaled by $(1 - y_i)^2$, meaning higher reliability levels (y_i closer to 1) lead to reduced variability.

Figures 8a (MWV(α)) and 9a (MV(α)) illustrate the order quantities as α increases from 0 to 1000 and finally ∞. In the risk-neutral setting ($\alpha = 0$), suppliers 5 and 2 are selected. As α increases, suppliers are added in the order: 4, 3, 8, 1, 10, 7, 9, 6. Note that this order matches the supplier ranking based solely on their unit *procurement cost*, aligning with the supplier activation observed in the risk-neutral setting. When the weighted variance objective is minimized ($\alpha = \infty$), all suppliers are activated. Ordering the suppliers based on their *order quantities* (from highest to lowest) yields the following sequence: 8, 1, 10, 7, 4, 6, 9, 3, 5, 2. This order is consistent with the increasing *variance of reliability factors* $\mathrm{Var}(R_i) = (1 - y_i)^2 p_i(1 - p_i)$. Finally, as anticipated, for the same value of α, the mean–variance setting produces a more risk-averse solution than the mean-weighted variance setting, activating all suppliers at smaller values of α. Figures 8b and 9b illustrate the relationship between the expected profit and standard deviation as the risk-aversion parameter α increases. The expected profit drops sharply for small increases in α, eventually turning negative when $\alpha > 0.2$. In contrast, the standard deviation remains relatively steady across larger values of α. Consequently, settings with high α are unlikely to be preferred by a rational decision-maker, as they lead to unacceptably low or even negative expected profits while the risk stays relatively stable.

6.3 Sensitivity analysis

In this section, we conduct a sensitivity analysis on key parameters, such as stock-out cost and demand variance. We solve the mean-risk problems with risk weight $\alpha = 0.0005$. Results for the sensitivity analysis can be found in Table 6. We also examine scenarios where risk-neutral and risk-averse newsvendors order more or less.

Table 6 Sensitivity analysis for MV(0.0005), MWV(0.0005), and risk-neutral (E)

Setting	Exp. Profit	St Dev	5	2	4	3	8	1	10	7	9	6
(a) MV(0.0005)												
Base ($k = 15, \sigma = 130$)	4522	2288	151	100	143	0	0	0	0	0	0	0
$k = 0, \sigma = 130$	5677	1723	201	112	0	0	0	0	0	0	0	0
$k = 100, \sigma = 130$	3220	4327	173	120	342	0	0	0	0	0	0	0
$k = 15, \sigma = 10$	6886	1072	218	194	93	0	0	0	0	0	0	0
(b) MWV(0.0005)												
Base ($k = 15, \sigma = 130$)	5336	3244	352	33	0	0	0	0	0	0	0	0
$k = 0, \sigma = 130$	6030	2108	311	0	0	0	0	0	0	0	0	0
$k = 100, \sigma = 130$	3377	4372	190	128	316	0	0	0	0	0	0	0
$k = 15, \sigma = 10$	7639	3578	402	8	0	0	0	0	0	0	0	0
(c) Risk-neutral (E)												
Base ($k = 15, \sigma = 130$)	6516	4045	516	21	0	0	0	0	0	0	0	0
$k = 0, \sigma = 130$	7279	3503	487	0	0	0	0	0	0	0	0	0
$k = 100, \sigma = 130$	4534	5854	453	247	0	0	0	0	0	0	0	0
$k = 15, \sigma = 10$	7641	3555	404	9	0	0	0	0	0	0	0	0

In the base setting, the stockout cost is $k = 15$. We now evaluate two scenarios: $k = 0$ and $k = 100$. When there is no penalty for stockouts ($k = 0$), the mean-weighted variance objective results in selecting only the cheapest supplier. For the mean–variance objective, risk diversification remains relevant, leading to the selection of the two cheapest suppliers. The selected supplier set aligns with our claim that cost is prioritized first, followed by reliability, within the mean-risk framework. Next, we increase the stockout cost to $k = 100$ while keeping all other parameters unchanged. With this high penalty for stockouts, both the mean-weighted variance and mean–variance objectives select the first three suppliers, increasing the order quantity from Supplier 4. Consistent with earlier results, supplier elimination is primarily driven by cost.

We decrease the standard deviation of demand to 10, resulting in an almost deterministic demand while keeping all other parameters unchanged. For the mean-weighted variance objective, only the two cheapest suppliers are selected, with the majority of the order placed with Supplier 5. This solution closely resembles the risk-neutral result, indicating that when demand uncertainty is minimal, risk diversification is no longer the dominant strategy. Instead, ordering primarily from the cheapest suppliers becomes more favorable. In contrast, for the mean–variance objective, risk diversification remains important, as the three cheapest suppliers are selected, with a considerable order quantity allocated to Supplier 4. While the total order quantity in the risk-neutral case decreases with lower demand risk, it increases for the Mean–variance objective.

According to Ray and Jenamani (2016a), a risk-averse decision-maker orders less than a risk-neutral one under supplier risk. Fleuren (2025) specifies conditions under which this holds true but does not consider supplier unreliability. Our numerical experiments reveal that the claim of consistent lower ordering by risk-averse decision-makers cannot be generalized. Detailed experimental results are provided in Supplementary Materials B.7.

7 Conclusion

This study provides a comparative analysis of supplier selection under *risk-neutral* and *risk-averse* objectives, with key contributions to both theoretical insights and computational efficiency.

In the *risk-neutral setting*, we address the limitations of the procedure proposed by Ray and Jenamani (2016b), which orders suppliers based on their *effective cost* and solves incrementally growing subsets of suppliers using a commercial nonlinear solver. We prove that this approach does not guarantee optimality. Instead, suppliers must be ordered by their *unit procurement cost* to achieve the optimal solution. This finding aligns with the *cost-first, reliability-second* principle observed in the risk-neutral literature and constitutes the first formal proof of this result under the FDG framework. One significant difference compared to the AON setting (as discussed by Mohammadivojdan et al. 2022) lies in the behavior of the service level. In particular, in the single-sourcing FDG setting, the service level can be non-monotone in the disruption probability, exhibiting a decrease–then–increase pattern: we prove this

unimodality analytically for Uniform demand and observe the same qualitative behavior numerically for Normal demand, with additional robustness results for other distributions reported in the Supplementary Materials B.4.

For the *risk-averse setting*, we extend the analysis to the *mean–variance* framework, as studied in Ray and Jenamani (2016a). We demonstrate that Ray and Jenamani (2016a) only partially included the variance terms in their formulation, leading to a modified risk measure, which we define as the *weighted variance*. We identified scenarios where minimizing variance provides additional benefits, particularly in settings with high supplier reliability variances. This is especially relevant for AON suppliers with moderate reliability. We further analyze supplier activation patterns as risk aversion increases. For *mean-risk* problems, suppliers are activated in an order that matches increasing procurement costs. When the decision-maker minimizes the weighted variance, we analytically demonstrate that the optimal solution retains all suppliers for a broad class of demand distributions. Furthermore, in the numerical tests, we observe that order quantities exhibit an inverse relationship with the variance of reliability factors. We observe similar results for the mean–variance objective as well. Finally, we observe that a risk-averse decision-maker may order more than a risk-neutral one when suppliers face risk and stockout costs are significant. In terms of computational efficiency, our algorithm utilizes the problem's *optimality conditions*, eliminating the need for solving incrementally growing subsets of suppliers.

Several avenues remain for future research. First, we assume that suppliers provide fixed fractional delivery guarantees, but it is worthwhile to study coordination mechanisms to ensure acceptable levels for both parties. Second, exploring more complex supplier behavior, such as capacity constraints, correlated disruptions, or strategic interactions, could provide a more realistic perspective on purchasing decisions.

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